Case Study 2 Project Proposal

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Load Packages

```
library(tidyverse)
```

Project choice # (1-5 based on Canvas descriptions): #4 Sports Reference

Dataset(s) within selected source, if applicable: Basketball

External data, if applicable:

Brief description: Player Stats by Season & All Pro Selections

The primary dataset would be a season-long aggregation of a player's box score. Advanced metrics like win-shares can be calculated from these statistics. In addition, there is a separate data set that contains All Pro selections for a season.

The other research question would use season-long team statistics (same as player statistics aggregated on a team level).

Research question 1: Can box scores and advanced metrics successfully predict All Pro selections using a generalized linear mixed effects model?

- Outcome variable (include the name/description and type of variable): All Pro Selection (Binary - Yes/No)

Research question 2 (optional): Can team statistics predict playoff teams, and if so how far in advance?

- Outcome variable (include the name/description and type of variable): Playoff Entrance (Binary - Yes/No)

Load the data and provide a `glimpse()`:

```r
library(tidyverse)
library(stringr)

player = read_csv("player_24-25.csv") |>
  mutate(Player = str_squish(Player))

all_pro = read_csv("all-pro_24-25.csv") |>
  pivot_longer(
    cols = everything(),
    names_to = "All-Pro",
    values_to = "Player") |>
  mutate(Player = str_squish(Player))

player = player |>
  left_join(all_pro, by = join_by(Player == Player))

glimpse(player)
```

```
Rows: 735
Columns: 34
$ Rk                  <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 12, 12, 13,~
$ Player              <chr> "Shai Gilgeous-Alexander", "Anthony Edwards", "Nik~
$ Age                 <dbl> 26, 23, 29, 30, 26, 28, 26, 25, 23, 35, 29, 29, 29~
$ Team                <chr> "OKC", "MIN", "DEN", "MIL", "BOS", "PHO", "ATL", "~
$ Pos                 <chr> "PG", "SG", "C", "PF", "PF", "SG", "PG", "SG", "PG~
$ G                   <dbl> 76, 79, 70, 67, 72, 75, 76, 77, 70, 79, 72, 74, 42~
$ GS                  <dbl> 76, 79, 70, 67, 72, 75, 76, 77, 70, 79, 72, 74, 42~
$ MP                  <dbl> 2598, 2871, 2571, 2289, 2624, 2795, 2739, 2725, 24~
$ FG                  <dbl> 860, 721, 786, 793, 662, 654, 566, 651, 684, 531, ~
$ FGA                 <dbl> 1656, 1612, 1364, 1319, 1465, 1420, 1376, 1378, 14~
$ `FG%`               <dbl> 0.519, 0.447, 0.576, 0.601, 0.452, 0.461, 0.411, 0~
$ `3P`                <dbl> 163, 320, 138, 14, 250, 183, 218, 251, 149, 235, 1~
$ `3PA`               <dbl> 435, 811, 331, 63, 728, 551, 642, 670, 418, 668, 3~
$ `3P%`               <dbl> 0.375, 0.395, 0.417, 0.222, 0.343, 0.332, 0.340, 0~
$ `2P`                <dbl> 697, 401, 648, 779, 412, 471, 348, 400, 535, 296, ~
$ `2PA`               <dbl> 1221, 801, 1033, 1256, 737, 869, 734, 708, 1039, 6~
$ `2P%`               <dbl> 0.571, 0.501, 0.627, 0.620, 0.559, 0.542, 0.474, 0~
$ `eFG%`              <dbl> 0.569, 0.547, 0.627, 0.607, 0.537, 0.525, 0.491, 0~
$ FT                  <dbl> 601, 415, 361, 436, 358, 432, 491, 287, 313, 505, ~
$ FTA                 <dbl> 669, 496, 451, 707, 440, 483, 561, 327, 370, 578, ~
$ `FT%`               <dbl> 0.898, 0.837, 0.800, 0.617, 0.814, 0.894, 0.875, 0~
$ ORB                 <dbl> 67, 61, 200, 147, 48, 76, 36, 35, 56, 55, 212, 22,~
```

```
$ DRB                 <dbl> 312, 389, 692, 651, 575, 229, 200, 364, 369, 401, ~
$ TRB                 <dbl> 379, 450, 892, 798, 623, 305, 236, 399, 425, 456, ~
$ AST                 <dbl> 486, 359, 716, 433, 431, 529, 880, 424, 638, 687, ~
$ STL                 <dbl> 131, 91, 127, 58, 76, 67, 91, 69, 71, 118, 71, 58,~
$ BLK                 <dbl> 77, 51, 45, 78, 38, 16, 12, 17, 53, 55, 50, 12, 10~
$ TOV                 <dbl> 183, 249, 230, 206, 209, 220, 355, 198, 309, 341, ~
$ PF                  <dbl> 164, 150, 160, 155, 157, 198, 145, 85, 195, 162, 2~
$ PTS                 <dbl> 2484, 2177, 2071, 2036, 1932, 1923, 1841, 1840, 18~
$ `Trp-Dbl`           <dbl> 0, 0, 34, 11, 2, 0, 0, 0, 9, 3, 1, 0, 0, 0, 0, 0, ~
$ Awards              <chr> "MVP-1DPOY-10CPOY-8ASNBA1", "MVP-7CPOY-3ASNBA2", "~
$ `Player-additional` <chr> "gilgesh01", "edwaran01", "jokicni01", "antetgi01"~
$ `All-Pro`           <chr> "1st Team", "2nd Team", "1st Team", "1st Team", "1~
```